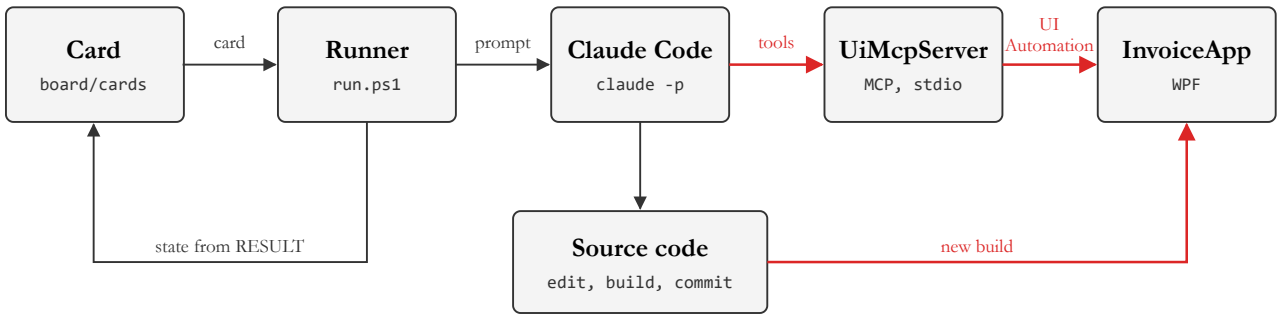
How to let Claude Code fix and test bugs in a WPF application

October 2026, Jürgen Bäurle

In September 2026 I wrote CardsToLoop, a small demo of a card-driven development loop for a WPF application. A Kanban card describes a bug fix or a new feature, and Claude Code implements it. For a desktop application a successful build proves nothing. The agent has to start the application, use it and check the result itself.

This article shows how to connect Claude Code with a small MCP server for Windows UI Automation and a PowerShell runner. The runner gives the agent one card, and the agent works on it until it commits the change. It is the same pattern as in the agentic UI test system for the Peakboard Designer, in about 700 lines.

The diagram below shows the parts and the way of a card through the loop. The red arrows mark the live test. The agent drives the application through the MCP server and checks every new build again.



The application and the cards

The demo has four parts:

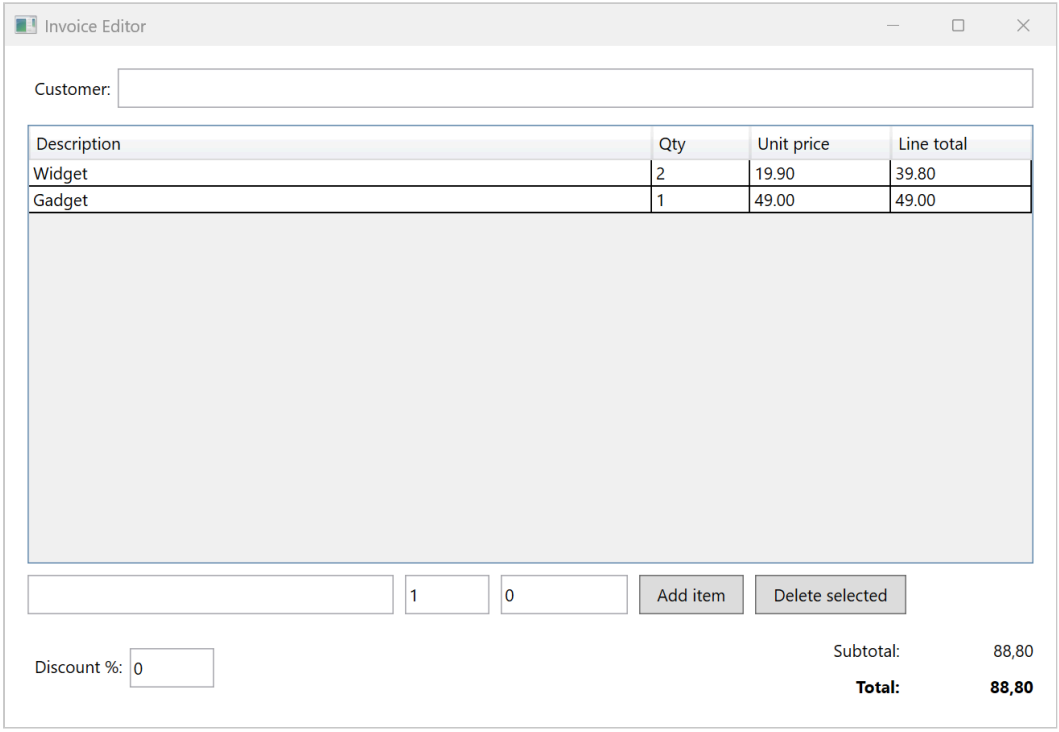
- InvoiceApp, a small WPF invoice editor with three deliberate gaps
- UiMcpServer, an MCP server with eight tools for Windows UI Automation
- three cards, one Markdown file per card
- run.ps1, the runner that starts Claude Code for one card

The repository has the following structure:

```
.claude/
  rules/            style rules for the code and the comments
  settings.json     permissions for the non-interactive run
.mcp.json           start command of the MCP server
board/
  cards/            one Markdown file per card
  run.ps1           the runner
  runs/             one log per run
src/
  InvoiceApp/       the WPF application
  UiMcpServer/      the MCP server
```

Claude Code reads the rules in .claude/rules on every run, so the agent writes code in the style of the project.

The InvoiceApp is a simplified example with a simple user interface. It only exists to show the principle of the technique. The screenshot below shows the InvoiceApp after the start, with two sample items:



The grid holds the items. Below it are the fields for a new item, the Discount % field and the totals.

Every control of the InvoiceApp has an AutomationId. An AutomationId is a fixed name of a control that UI Automation reports to other programs. The agent uses it to find the Discount % text box or the total without a screenshot. The following XAML shows the Discount % text box:

```
<TextBox x:Name="DiscountTextBox"
         AutomationProperties.AutomationId="DiscountTextBox"
         Width="60" Height="28" VerticalContentAlignment="Center" Text="0"
         TextChanged="DiscountTextBox_TextChanged"/>
```

The attribute AutomationProperties.AutomationId sets the AutomationId. The agent passes this value to the tools of the MCP server to address the text box.

A card is a Markdown file with a title, a state, a description and acceptance criteria. Acceptance criteria are testable conditions that the result has to meet before the card is done. The state moves from New to "In Work" and then to Done or Failed. See the example in the section "A bug fix card" below.

The repository contains three cards:

- The total ignores the discount, a bug
- Deleting an item should ask first, an improvement
- Validate quantity and price before adding an item, an improvement

This article shows the run of the bug card.

The MCP server

The Model Context Protocol (MCP) defines how an agent calls the tools of another process. UiMcpServer is a .NET 8 console application with the ModelContextProtocol package in version 2.2.0. It offers the following tools:

- **ui_list_windows** for the visible top-level windows
- **ui_find_elements** for all elements of a window with an AutomationId or a name
- **ui_click** for a click on an element
- **ui_set_text** for writing the text of a text box
- **ui_get_text** for reading the text of an element
- **ui_select_row** for the selection of a row in a list or grid
- **ui_wait_for** for waiting until an element exists
- **ui_screenshot** for a PNG screenshot of the window

Every tool addresses a window by a part of its title and an element by its AutomationId. If no element has this AutomationId, the tool searches by name. So the buttons of a system dialog stay reachable, because they have a caption but no AutomationId.

The following code shows how a tool finds the element:

```
private static AutomationElement FindTarget(AutomationElement window, string target)
{
    var byId = window.FindFirst(TreeScope.Descendants,
        new PropertyCondition(AutomationElement.AutomationIdProperty, target));
    if (byId is not null)
    {
        return byId;
    }
    var byName = window.FindFirst(TreeScope.Descendants,
        new PropertyCondition(AutomationElement.NameProperty, target));
    if (byName is not null)
    {
        return byName;
    }
    throw new McpException($"No element with AutomationId or Name '{target}'.");
}
```

FindFirst with TreeScope.Descendants searches the whole tree below the window. The first search uses the AutomationId, the second the Name property. If both searches fail, the tool throws an McpException. The SDK returns its message to the agent as an error result, for example "An error occurred invoking 'ui_set_text': No element with AutomationId or Name 'NoSuchBox'." With any other exception the agent only gets the first part and does not learn the reason.

Before the agent sets a value, it calls ui_find_elements. For the InvoiceApp the tool returns 60 elements, including the title bar and the parts of the grid. The following excerpt shows the three elements of the totals area:

```
{ "controlType": "Edit", "automationId": "DiscountTextBox", "name": "", "value": "0" }
{ "controlType": "Text", "automationId": "SubtotalText", "name": "88,80", "value": null }
{ "controlType": "Text", "automationId": "TotalText", "name": "88,80", "value": null }
```

The text of a text box is in value, the text of a TextBlock is in name. The totals show 88,80, the sum of the two sample items.

The following code shows the tool that sets the text of a text box:

```
[McpServerTool(Name = "ui_set_text")]
[Description("Replaces the text of a text box via the Value pattern.")]
public static string SetText(
    [Description("Title fragment of the window")] string windowTitle,
    [Description("AutomationId or Name of the element")] string target,
    [Description("The new text")] string text)
{
    var element = FindTarget(FindWindow(windowTitle), target);
    if (!element.TryGetCurrentPattern(ValuePattern.Pattern, out var value))
    {
        throw new McpException($"'{target}' does not support the Value pattern.");
    }
    ((ValuePattern)value).SetValue(text);
    return Ok($"set '{target}' to \"{text}\"");
}
```

The McpServerTool attribute registers the method under the name ui_set_text. The Description attributes are part of the prompt. Claude Code sends them to the model, and the model decides from them which tool to call with which values. The Value pattern is the UI Automation interface for controls with a text value. The click tool tries the Invoke, SelectionItem and Toggle patterns first and sends a real mouse click only as the last step.

A file called .mcp.json in the repository tells Claude Code how to start the server:

```
{
  "mcpServers": {
    "ui": {
      "command": "src/UiMcpServer/bin/Debug/net8.0-windows/UiMcpServer.exe",
      "args": []
    }
  }
}
```

Claude Code starts the server as a child process and talks to it over standard input and output. Each message is one line of JSON-RPC. The following two lines show a call of ui_set_text and the answer of the server:

```
{"jsonrpc":"2.0","id":2,"method":"tools/call","params":{"name":"ui_set_text","arguments":
{"windowTitle":"Invoice Editor","target":"DiscountTextBox","text":"10"}}}
{"jsonrpc":"2.0","id":2,"result":{"content":[{"type":"text","text":"{\"ok\":true,\"message\":\"set
'DiscountTextBox' to \\\"10\\\"\"}"}]}}
```

Before the first call Claude Code sends initialize with the protocol version and then tools/list. The answer to tools/list contains the name, the description and a JSON schema of the parameters of every tool. The SDK creates the schema from the method and its attributes. The following excerpt shows the entry of ui_set_text:

```
{
  "name": "ui_set_text",
  "description": "Replaces the text of a text box via the Value pattern.",
  "inputSchema": {
    "type": "object",
    "properties": {
      "windowTitle": { "description": "Title fragment of the window", "type": "string" },
      "target": { "description": "AutomationId or Name of the element", "type": "string" },
      "text": { "description": "The new text", "type": "string" }
    },
    "required": [ "windowTitle", "target", "text" ]
  }
}
```

The descriptions of the parameters come from the Description attributes in the code above. Without them the model would only see the names windowTitle, target and text.

The runner

The runner sets the card to "In Work", builds the prompt and starts Claude Code in non-interactive mode. The prompt contains the card, the paths of the project and five steps. The following text shows the template, and the runner inserts the card at $cardText:

```
You are working one card of a card-driven development loop in this repository.

THE CARD
--------
$cardText

THE PRODUCT
-----------
- WPF app: src/InvoiceApp (project src/InvoiceApp/InvoiceApp.csproj)
- Built exe: src/InvoiceApp/bin/Debug/net8.0-windows/InvoiceApp.exe
- Every control has an AutomationId; discover them with ui_find_elements.

HOW TO WORK
-----------
1. Start the app (Start-Process the exe), reproduce or inspect the behavior the
   card describes using the ui_* MCP tools. Take a screenshot as evidence.
2. Implement the change in src/InvoiceApp. Keep it minimal and in the style of
   the existing code.
3. Stop the app (taskkill /IM InvoiceApp.exe /F) BEFORE rebuilding - the build
   fails while the exe runs. Rebuild with: dotnet build src/InvoiceApp/InvoiceApp.csproj
4. Start the app again and verify every acceptance criterion of the card live
   through the ui_* tools: change values, click, read the results back.
   Take an after screenshot. A build that compiles proves nothing.
5. Stop the app. Commit ONLY the source changes (git add the changed files
   under src/, never git add -A) with a message whose first line is:
   card: <the card title>
   and whose body says in two or three sentences what was changed and how it
   was verified live.

RULES
-----
- Do not touch board/, README.md or anything outside src/ and the commit.
- If an acceptance criterion cannot be met, do not commit half work.

THE LAST LINE of your final message must be exactly one of:
RESULT: done
RESULT: failed: <one-line reason>
```

The fourth step holds the key sentence of the prompt: "A build that compiles proves nothing." The rules keep the agent away from the board and stop a commit of half work. The following code shows the call and the evaluation:

```
$output = claude -p $prompt --mcp-config .mcp.json --strict-mcp-config 2>&1
$output | Set-Content $logPath -Encoding UTF8

$resultLine = @($output | Where-Object { $_ -match "^RESULT:" }) | Select-Object -Last 1
if ($resultLine -match "^RESULT: done")
{
    Set-CardState "Done"
}
else
{
    Set-CardState "Failed"
}
```

The -p flag runs Claude Code without its terminal interface and prints the last message. The --strict-mcp-config flag loads only the server from .mcp.json and no server from the user settings. The last line of the answer has to be **RESULT: done** or **RESULT: failed** with a reason. The runner moves the card depending on this line and writes the output to a log file.

The following function changes the state in the front matter of the card:

```
function Set-CardState([string]$state)
{
    $script:cardText = $script:cardText -replace "(?m)^state: .*$", "state: $state"
    Set-Content $card.FullName $script:cardText -Encoding UTF8 -NoNewline
    Write-Host "card '$($card.Name)' -> $state"
}
```

The function replaces the state line and writes the file back. During the run the card shows "In Work", after the run Done or Failed.

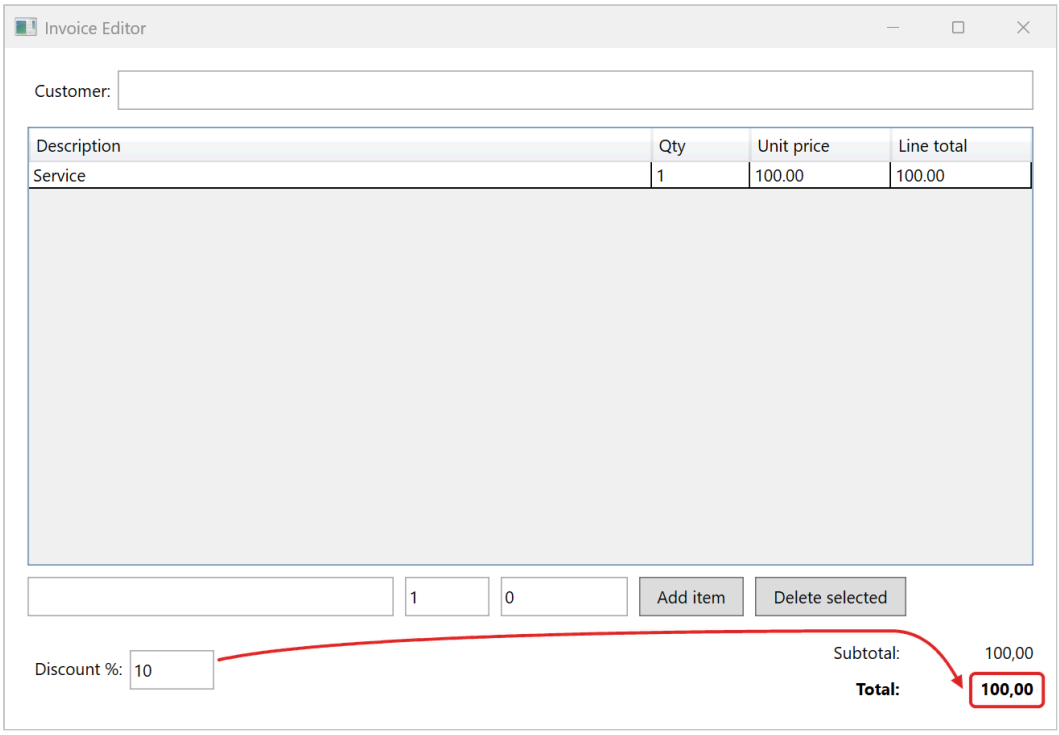
In non-interactive mode Claude Code cannot ask for permission. The file .claude/settings.json in the repository allows the file tools, Bash and the UI tools. Claude Code reads it only once the folder is trusted. My first run failed at this point. Claude Code ignored the permissions with a warning on stderr, and Windows PowerShell stopped the script because of this warning. Bash without restriction is fine on a development machine, not on a shared one. The following code shows the file:

```
{
  "permissions": {
    "allow": [
      "Read",
      "Glob",
      "Grep",
      "Edit",
      "Write",
      "Bash",
      "mcp__ui",
      "mcp__ui__*"
    ]
  }
}
```

The entry mcp__ui__* allows all tools of the server called ui in .mcp.json.

A bug fix card

One of the three cards describes a bug. The InvoiceApp has a Discount % field, but the total ignores it. With one item of 100.00 and a discount of 10 the total stays at 100,00 (marked in red in the screenshot below).



The following card describes the bug and the expected result:

```
---
title: The total ignores the discount
state: New
category: Bug
---

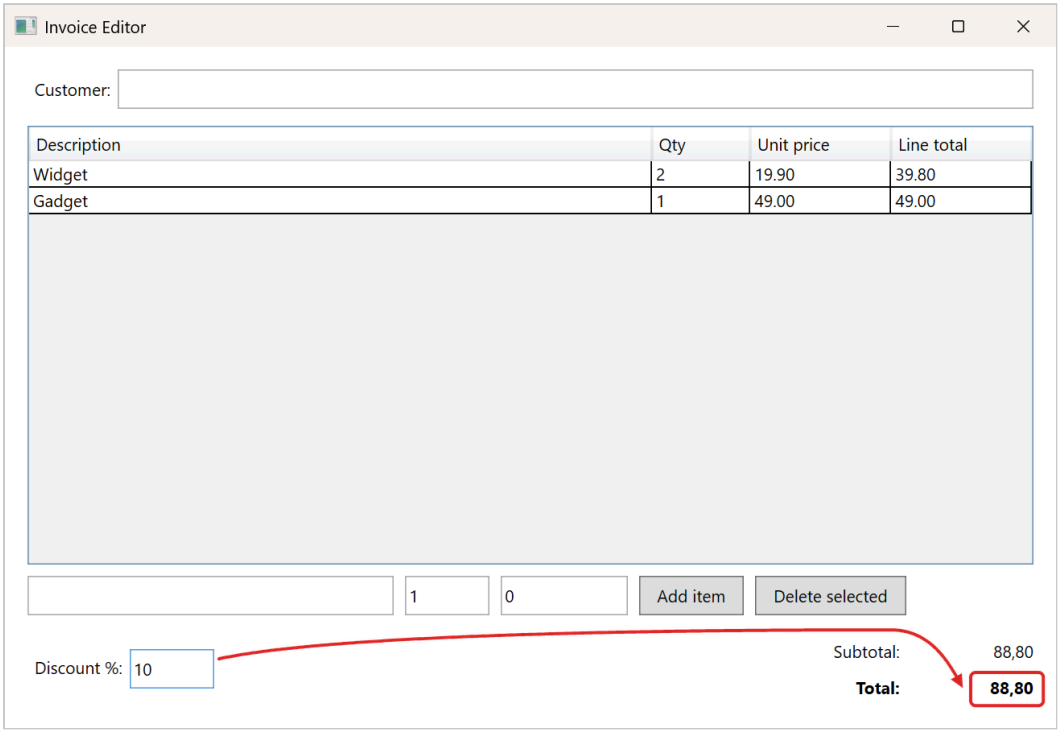
The totals panel has a "Discount %" field. Typing a discount changes nothing:
the total always equals the subtotal.

**Acceptance criteria**

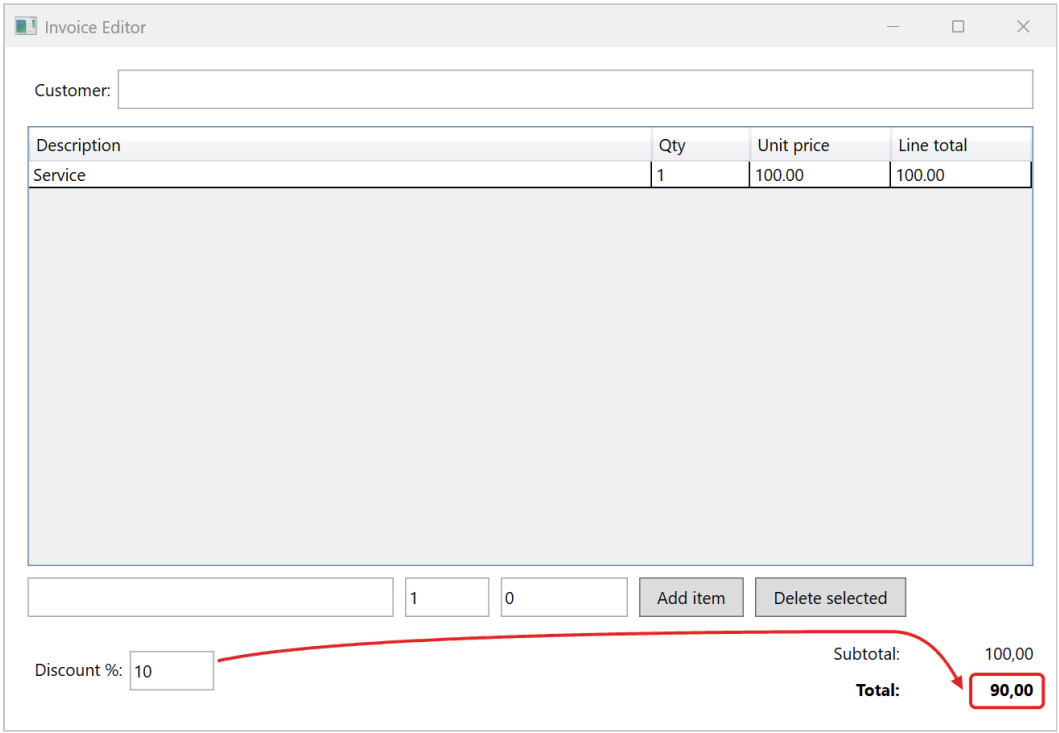
- With items worth 100.00 and a discount of 10, the total shows 90.00.
- An empty or non-numeric discount counts as 0, not as an error.
- The subtotal stays undiscounted.
```

Each acceptance criterion names a value to enter and a value to read back.

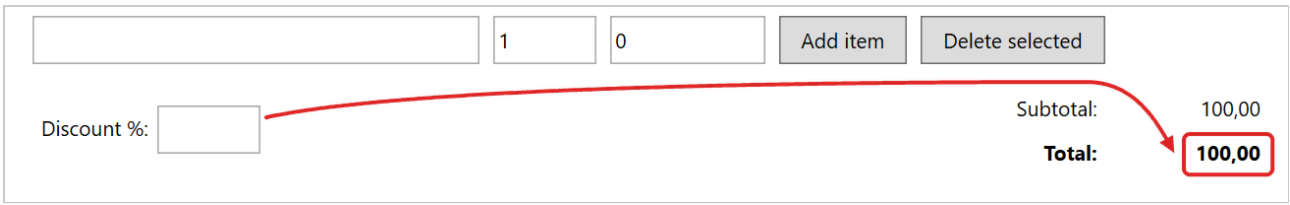
The agent first reproduced the bug with the two sample items. The screenshot below is the one the agent took with ui_screenshot, with the red marks added. A discount of 10 leaves the total at 88,80.

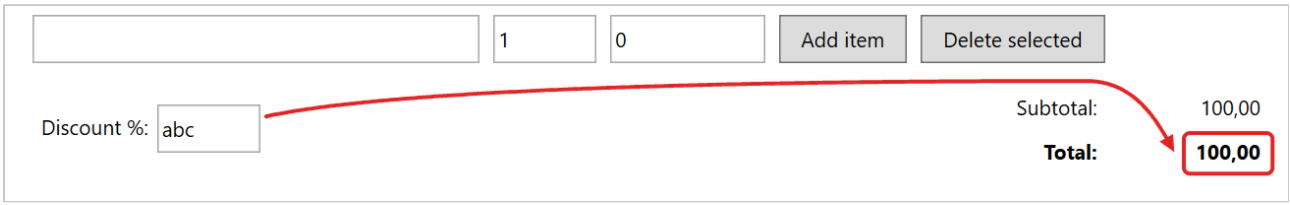


Then it changed the code, built the application again and checked the three criteria. The next screenshot shows the result. With one item of 100.00 and a discount of 10 the total is now 90,00 (marked in red).



The agent checked the other two criteria in the same way. The next two screenshots repeat these checks with an empty discount and with the text abc. They show only the lower part of the window. In both cases the total is 100,00, and no error message appears.





The following diff shows the change in a method called RecalculateTotals:

```
 SubtotalText.Text = subtotal.ToString("N2", CultureInfo.CurrentCulture);
-// NOTE: The discount is read but never applied on purpose - this is the bug the
-// first card describes, and the agent run fixes it.
-TotalText.Text = subtotal.ToString("N2", CultureInfo.CurrentCulture);
+// An empty or non-numeric discount counts as 0.
+var discountPercent = decimal.TryParse(DiscountTextBox.Text, NumberStyles.Number,
+    CultureInfo.CurrentCulture, out var d) ? d : 0m;
+var total = subtotal - subtotal * discountPercent / 100m;
+TotalText.Text = total.ToString("N2", CultureInfo.CurrentCulture);
```

The new code reads the discount with the culture of the machine and counts an empty or invalid value as 0. Then it subtracts the discount from the subtotal. The commit contains only MainWindow.xaml.cs, with the following message:

```
card: The total ignores the discount

RecalculateTotals now parses the Discount % box with the current culture and
subtracts that share from the subtotal; an empty or non-numeric value counts
as 0. The subtotal line is unchanged. Verified live through UI Automation:
with one item of 100.00 a discount of 10 shows total 90.00, an empty box and
"abc" both show 100.00 with no error dialog, and the subtotal stayed at
100.00 throughout.
```

The last message of the agent in the log reads as follows:

```
Committed as 26b1890. Only the source file went in; the card file edit from the
runner stays in the working tree.

Change: `MainWindow.xaml.cs:68` now parses the discount box with the current
culture, falls back to 0 for empty or non-numeric input, and sets the total to
subtotal minus that share.

Verified live on the rebuilt exe through the UI tools:
- Before the fix, discount 10 on the default items left the total at 88,80,
  equal to the subtotal.
- One item of 100.00 with discount 10 shows total 90,00.
- Empty discount and "abc" both show total 100,00, no error dialog appeared.
- Subtotal stayed at 100,00 in every case.

Screenshots are in the temp folder as total-ignores-the-discount-before.png and
-after.png, kept out of board/ per the rules. The app is stopped.

RESULT: done
```

The runner reads only the last line and sets the card to Done. In the repository the tag article-2-run marks the state after this run. The main branch keeps the bug, so the card can run again.

Limits and next steps

The run also shows a limit. The InvoiceApp formats the totals with the German culture of the machine and the grid with English numbers. The card asks for 90.00, the agent read 90,00 and accepted it without a note.

This article does not introduce MCP or UI Automation in detail. There are other ways to let an agent see an application. A screenshot and a model with image input need no AutomationIds, but every step costs an image and values are read less exactly. For web applications Playwright offers an MCP server with the same kind of tools. UiMcpServer uses System.Windows.Automation, the managed UI Automation API of .NET. For applications with WinForms or Win32 controls, the library FlaUI with the newer COM interface of UI Automation is an alternative. Recorded scripts run faster and without a model, but they break when the user interface changes and cannot judge a result.

In the demo no human takes part between the card and the commit. A next version could add the following:

- a Questions state for unclear cards, with the questions and the answers in the card
- a Review state before the commit, with my feedback for a second round
- a check of the card without write access, before any code changes
- a project memory, in which the agent notes what it learned for the next card
- a Kanban board as a small web service

For questions and review the runner would continue the same Claude Code session with the --resume flag. So the agent would keep what it already knows about the application. Claude Code reads a CLAUDE.md in the repository at the start of every run. If the agent added what it learned there, for example that the totals use the German number format, the next card would start with this knowledge. These entries would need the same review as the code.

The Kanban board could show one column per state and start the runner for a card. For each run it would show the log, the diff and the screenshots and offer fields for answers and feedback. The cards would stay Markdown files in the repository, and the board would only read and write them. It could also work through the New column one card after another and send a notification when a card waits for me.

That's it.

If you have questions feel free to contact me via www.parago.de.

Links

CardsToLoop on GitHub: https://github.com/jbaurle/CardsToLoop
Model Context Protocol: https://modelcontextprotocol.io/
MCP C# SDK: https://github.com/modelcontextprotocol/csharp-sdk
Claude Code, run programmatically: https://code.claude.com/docs/en/headless
Claude Code, MCP: https://code.claude.com/docs/en/mcp
UI Automation overview: https://learn.microsoft.com/en-us/dotnet/framework/ui-automation/ui-automation-overview
FlaUI: https://github.com/FlaUI/FlaUI